PHIRMY

BUILD • SCALE • TOGETHER

Your website.
Ready for a conversation.

Phirmy: A Website-Grounded AI Assistant for Text, Voice, and Business Actions

Phirmy turns a business website and its documents into a source-grounded assistant. Visitors can type or speak, explore product and service information, and request supported business actions from one conversation.

The business connects its content, publishes an agent, and installs a widget script. The workspace brings together knowledge, agents, tools, conversations, leads, integrations, and usage.

Built on open technologies, with Sarvam conversational AI and a configurable chat-model layer.

DEPLOYED BETA / OPEN TECHNOLOGIES / STREAMING VOICE

phirmy.com | app.phirmy.com

The problem and our contribution

Problem statement

A website may already contain the answer, but visitors still have to search menus, read long pages, and wait for support. Businesses repeat the same explanations across product questions, policies, and service inquiries.

A useful assistant needs more than fluent text: it must find evidence, understand the current page, connect speech with business context, and keep actions within explicit permissions.

What already exists

Chatbase, Botpress, and Voiceflow already provide business-content assistants and deployment tools. Vapi and Retell provide voice-agent capabilities; LiveKit and Pipecat offer open-source realtime foundations. RAG, voice widgets, and model switching are established ideas.

What we are adding

Website-aware behavior	Current-page evidence, product information, and validated page navigation.
Shared text and voice context	One business knowledge and tool context across conversation modes.
Indic voice integration	Sarvam recognition, conversational replies, and speech generation connected through streaming interfaces.
A manageable business product	Knowledge publishing, agent configuration, tools, conversations, and operational controls in one workspace.

Who it is for

Small online retailers, service businesses, agencies, and developers who want to make website information conversational. Initial uses include FAQs, product discovery, policy guidance, consented lead capture, and configured scheduling assistance.

What we built and what we solved

Website-to-agent setup	URL onboarding connects a knowledge base, agent binding, and widget installation; workers crawl and index permitted content.
Grounded answers	Scoped retrieval, citations, structured product evidence, and current-page context connect responses to actual sources.
Portable widget	Custom branding, streaming text, voice controls, and same-tab conversation resume.
Streaming voice	Partial transcripts, incremental synthesis, interruption handling, playback clearing, connection reuse, and cleanup.
Controlled tools	Custom HTTP tools, schema checks, tenant permissions, confirmation flows, and stable execution identities.
Business workspace	Agent drafts/publishing, inbox, contacts/leads, catalog, appointments, integrations, team access, and developer APIs.

Engineering problems addressed

Versioned publishing keeps active conversations tied to consistent agent, knowledge, and tool revisions. Failed crawls preserve the last working snapshot. Website actions use known indexed destinations rather than model-generated JavaScript.

Recorded evidence

A real-site acceptance run indexed 48 pages, 83 passages, and 18 structured products. A separate live Sarvam check completed speech recognition, model reasoning, a read-only fixture tool, and spoken output. These are dated project checks, not universal coverage or customer-impact claims.

Technology stack

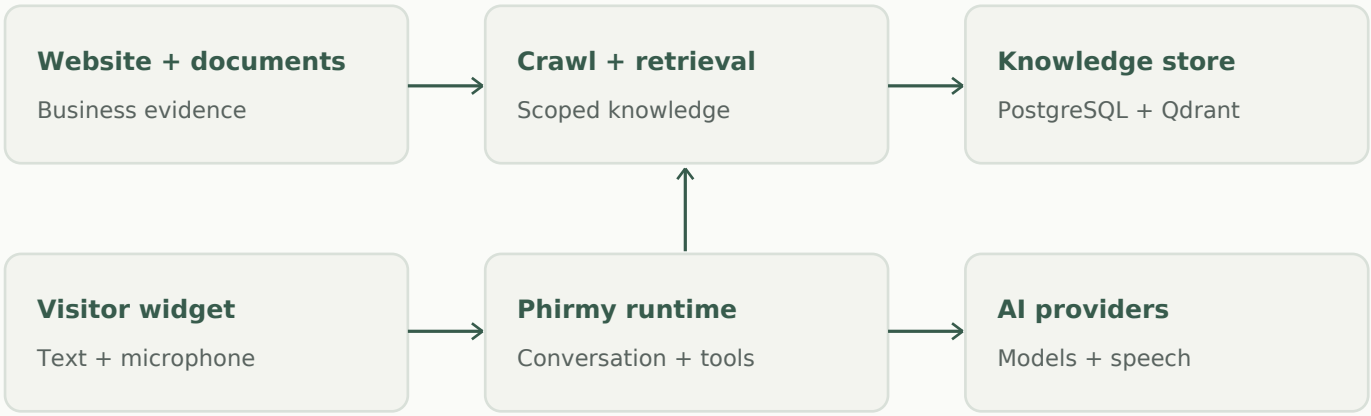
Frontend	Next.js, React, TypeScript, Lucide icons; embedded website widget and browser audio capture/playback.
Backend and APIs	Python, FastAPI, Uvicorn, Pydantic, SQLAlchemy; HTTP APIs, server-sent events, and WebSocket media.
Database and retrieval	PostgreSQL through Supabase for application records; Qdrant for knowledge retrieval; Redis for derived caching.
Content and storage	Guarded Playwright/Chromium rendering, HTML/document extraction, chunking, and private Azure object storage.
Text and embeddings	Configurable compatible chat-model profiles; current hosted GPT profile and text-embedding-3-small.
Speech recognition	Sarvam saaras:v4: realtime ASR, partial/final transcripts, and speech activity events.
Voice reasoning	Sarvam sarvam-105b-conversations: streamed conversational text and structured tool calls.
Speech synthesis	Sarvam bulbul:v4-flash: incremental TTS and streamed audio output.
Deployment and jobs	Docker Compose, Nginx, HTTPS, GitHub Actions; separate API and durable worker services, health checks, and scheduled backups.

Architecture choice

The browser handles interaction and audio. The backend controls retrieval, model calls, permissions, tools, and persistence. The worker handles long-running ingestion and background work.

Open-source frameworks and self-managed application services are combined with hosted model, speech, database, and storage integrations. A provider SDK or API is not itself proof of an open-source model licence.

Technical flow



Text conversation flow

Visitor question + current-page context -> widget session validation -> scoped knowledge retrieval -> selected chat model -> permitted tool/action loop -> persisted reply, citations, and streamed response.

Voice conversation flow

Microphone permission -> captured PCM audio -> backend media WebSocket -> Saaras v4 transcripts -> knowledge and conversation context -> Sarvam-105B Conversations + permitted tools -> Bulbul v4 Flash audio -> browser playback.

Knowledge ingestion flow

Website URL or document -> durable crawl/parse job -> safe extraction and chunking -> private object storage + source records -> embeddings and Qdrant index -> published knowledge snapshot -> session-scoped retrieval.

Tools execute only after application checks and any required confirmation. Interruptions clear old audio and cancel the prior producer. New knowledge is published through versioned snapshots; active sessions retain their pinned versions.

Open technologies and our USP

Open-source tools already used

Web experience	Next.js, React, TypeScript, and Lucide icons.
Application backend	Python, FastAPI, Uvicorn, Pydantic, and SQLAlchemy.
Data and retrieval	PostgreSQL, Qdrant, and Redis.
Content and operations	Playwright/Chromium, document parsers, Nginx, and containerized services.
Integrated open-weight model	Sarvam-105B Conversations; Sarvam lists Apache 2.0 for the 105B model family.

USP - Unique Selling Proposition

Phirmy brings website knowledge, page-aware assistance, and Indic voice into one configurable business assistant, with actions governed by the application.

Its strongest demonstration is the combination: evidence from the business website, shared text/voice context, published knowledge and tools, and control of the application runtime.

Model flexibility

Authorized users can add compatible HTTPS chat-model profiles with documented parameters and a streaming test, then select and publish a profile. The widget and business workflow can stay consistent as validated chat-model choices change.

The current Sarvam voice mode selects its own conversational model. A text-profile change does not replace ASR, TTS, or the voice LLM. Hosted APIs have separate terms; using open-source frameworks does not make every hosted component open source. The project repository is currently private.

Feasibility, impact, and next steps

Feasibility

Technical	A deployed beta already connects ingestion, retrieval, chat, voice orchestration, and tools using standard interfaces.
Operational	Separate API/worker services, durable jobs, health checks, versioned publishing, CI/CD, and scheduled backups support beta operation.
Adoption	The business installs a script and manages the assistant through a dashboard. Voice needs browser permission and a suitable secure host context.

Expected impact

Make business information easier to discover, reduce repetitive explanations, and help visitors move from a question to an explicit next step. Validate these aims through reviewed answer quality, task completion, real-device voice responsiveness, and user feedback.

What we are adding next

Real-microphone and multilingual evaluation; compatible NVIDIA Nemotron Ultra and GPT-OSS chat-profile experiments; a future NVIDIA Riva ASR adapter; selected real-account integration acceptance; and repeatable quality and capacity checks.

Nemotron/GPT-OSS and NVIDIA ASR are researched options, not completed Phirmy integrations. NVIDIA lists development APIs for Nemotron ASR Streaming, Parakeet, and Canary; access is subject to service limits. Riva needs an adapter. Model availability should be rechecked before a demo.

Three-minute demo

Show indexed website content. Ask a product or policy question and open its source. Demonstrate current-page guidance. Try a short voice exchange on a prepared device. Finish with the saved conversation, agent controls, and future model choices.

Presenter reference and sources

Closing pitch

Phirmy makes a business website something people can talk to. It combines grounded knowledge, streaming voice, useful tools, and configurable chat models in a deployed platform that businesses can manage around their own content.

Suggested slide sequence

Project title -> problem -> existing solutions -> our contribution -> what we built -> architecture -> open technologies -> USP -> feasibility and impact -> demo and future scope.

Official research sources

Sarvam 105B - model family and Apache 2.0 listing

Sarvam Saaras - recognition models

Sarvam Bulbul - speech generation

NVIDIA Nemotron 3 Ultra - candidate model

NVIDIA GPT-OSS-120B - model availability

NVIDIA Nemotron ASR Streaming

Chatbase - existing product capabilities

Botpress - website/document knowledge

Voiceflow - agent building and deployment

Vapi - modular voice architecture

Retell - text and voice website widgets

LiveKit Agents - open-source voice framework

Pipecat - open-source realtime framework

Project evidence is recorded in docs/AURELY_CRAWL_TEST.md, docs/SARVAM_VOICE.md, and docs/PRODUCTION_DEPLOYMENT.md. Physical-microphone acceptance and real-account workflows remain scoped acceptance tasks. No new provider requests were made to prepare this brief.